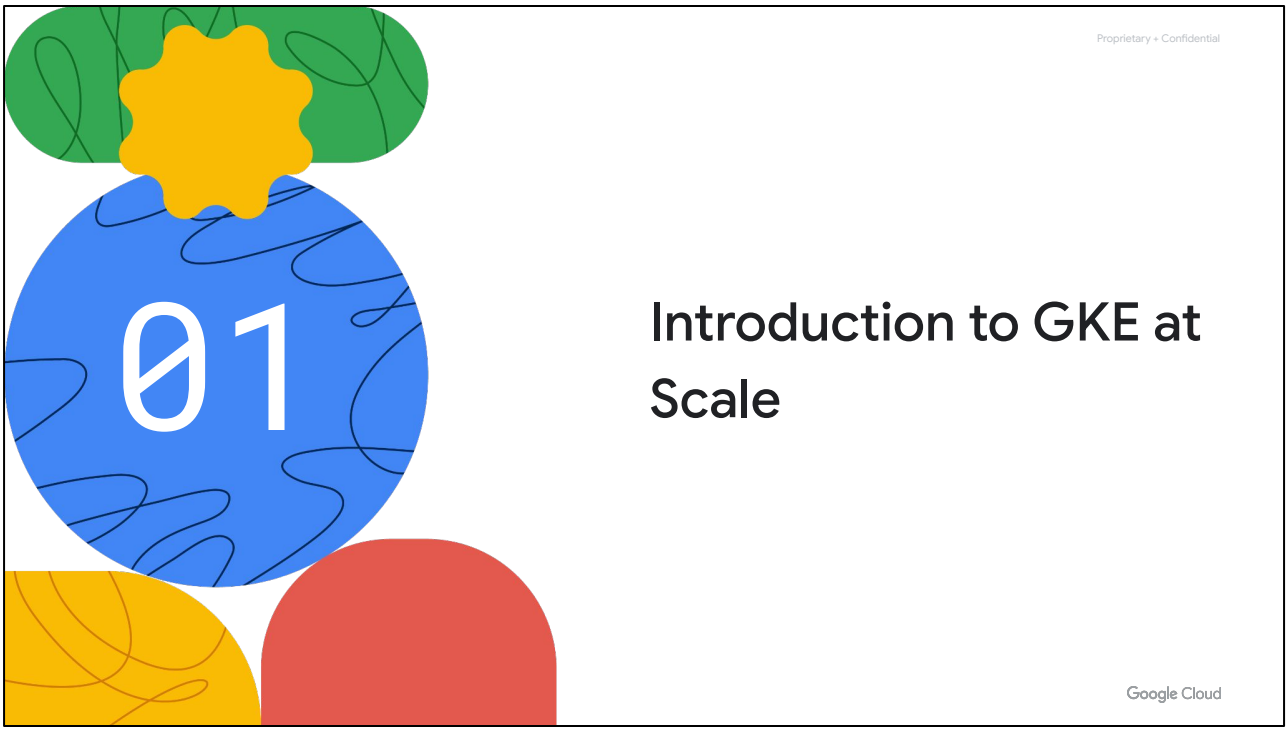

Kubernetes is the standard for container orchestration. But managing Kubernetes at scale, across multiple teams and with varying workloads, brings its own set of challenges. As organizations scale, many shift to a multi-cloud model to support their growing demands. This is where GKE comes in.

This course on managing GKE at scale is designed to help organizations like yours run critical workloads across clouds in a secure, scalable, and highly available manner. It builds upon the foundation of GKE, adding advanced features and capabilities to simplify management, enhance security, and boost operational efficiency.

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Introduction to GKE at scale

01 Multi-cloud and multi-cluster overview

02 Introduction to GKE Fleets

03 Sameness and trust

04 GKE fleet management features

Google Cloud

In the first section of this course titled, “Introduction to GKE at Scale”, you’ll learn to:

- Recognize the challenges of designing and building multi-environment solutions.
- Explain how GKE uses fleets to streamline operations.
- Describe the concepts of sameness and trust and use them to manage fleets.
- Identify the features and components used to manage GKE fleets.

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Google Cloud

In today's cloud-driven world, many enterprise organizations are adopting hybrid and multi-cloud approaches. According to a Flexera study, the average enterprise uses 2.6 cloud providers.

Why are enterprise organizations transitioning to hybrid and multi-cloud strategies? The shift is driven by a variety of factors, depending on the scale and complexity of the organization. Let's explore some of the key factors.

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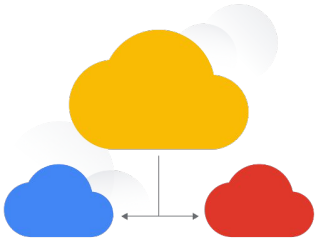
Multi-cloud needs



- Meet compliance requirements
- Minimize risk
- Increase resilience
- Leverage existing cloud commitments

Google Cloud

First, multiple cloud providers can help organizations **meet compliance requirements**. For example, companies with customers in the public sector are required to use a multi-cloud approach to meet data sovereignty and security regulations.

Adopting a multi-cloud strategy can also help **minimize risk** by giving companies more leverage when negotiating prices with cloud vendors.

When multiple providers are involved, Enterprise organizations also gain **resilience**. For example, in a situation where a provider does not fulfill their availability guarantees or a major disaster recovery is needed, a secondary provider can help with backup.

Finally, mergers and acquisitions often result in a multi-cloud environment, as companies **inherit existing cloud commitments** from the businesses they acquire.

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Introduction to GKE

01 Multi-cloud and multi-cluster overview

02 Introduction to GKE Fleets

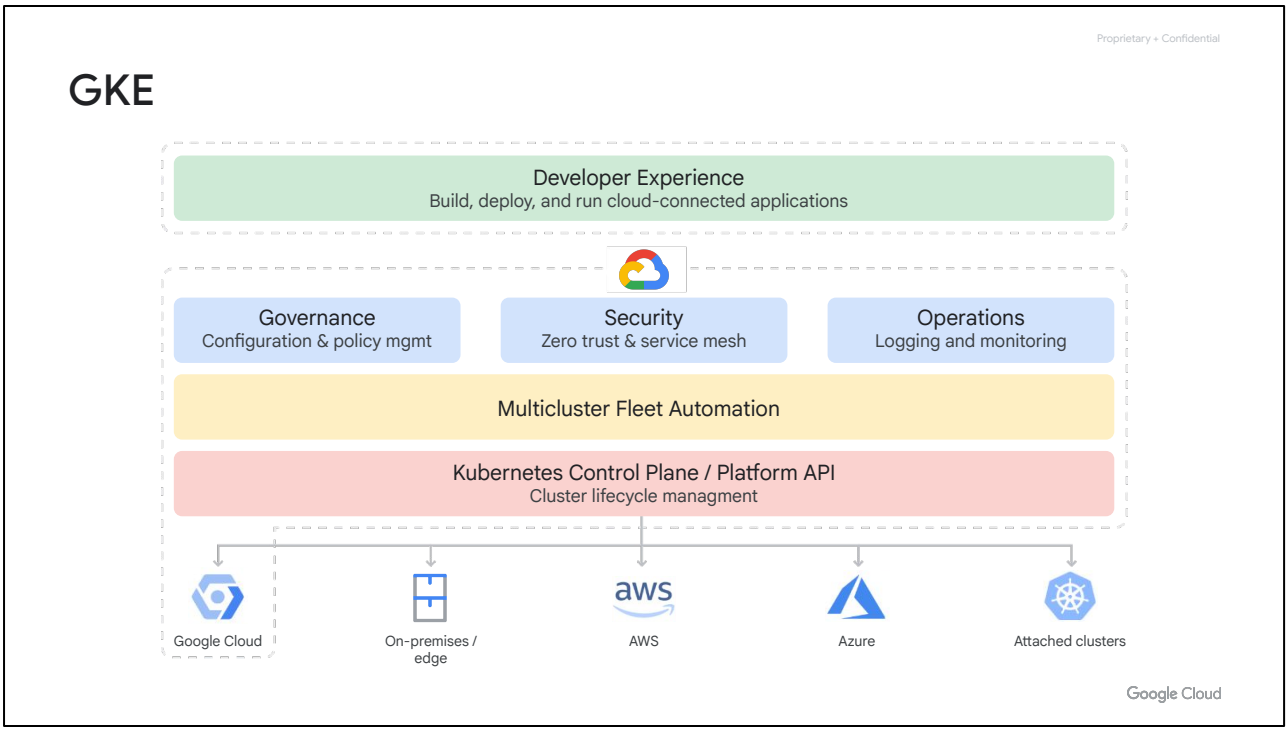
03 Sameness and trust

04 GKE features

Google Cloud

Google Kubernetes Engine, or GKE, is a powerful container orchestration tool that helps you manage your clusters in Google Cloud. But what if you have many clusters spread across different projects? Or maybe your clusters are hosted outside Google Cloud, either on-premises, in other other public clouds, or both.

How do you successfully manage all this infrastructure? Well, this is where GKE can help.



GKE offers extended capabilities like multi-environment deployments, consistent data security, and multi-cluster capability.

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Govern, manage, and operate containerized workloads at scale

Modernize your existing applications and infrastructure without needing to completely rebuild them.

Use a unified cloud operating model to create, update, and optimize container clusters regardless of where they're located.

Scale large multi-cluster applications with consistent security, configuration, and service management.

Enforce consistent governance and security from a unified control plane.

Google Cloud

GKE provides tools and features to help govern, manage, and operate containerized workloads at enterprise scale, enabling you to adopt the same best practices and principles that Google uses to run their services.

You can use GKE to:

- Modernize your existing applications and infrastructure without needing to completely rebuild them.
- Use a unified cloud operating model to create, update, and optimize container clusters regardless of where they're located.
- Scale large multi-cluster applications as fleets—logical groupings of similar environments—with consistent security, configuration, and service management.
- And enforce consistent governance and security from a unified control plane.

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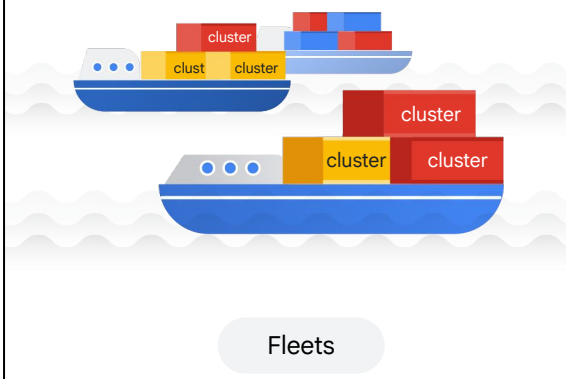
GKE uses fleets to streamline operations



- A logical grouping of Kubernetes clusters that can be managed together
- Can include Kubernetes clusters on Google Cloud, on-premises, or on another cloud provider

Google Cloud

GKE uses fleets to streamline operations and provide a unified platform for managing multiple clusters.

A fleet is a logical grouping of Kubernetes clusters that can be managed together. A fleet can include any Kubernetes cluster, whether it's running on Google Cloud, on-premises, or on another public cloud like AWS or Azure. This course focuses on GKE clusters running on Google Cloud.

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GKE expands beyond Kubernetes

Configuration and policy management

Scale your workloads by automatically adding and updating identical configurations, features, and security policies across your fleet.

Cloud Service Mesh

Application security, networking, and observability for microservice-based applications.

Observability

Monitor and troubleshoot your clusters and applications.

Team management

Provide team access and visibility to infrastructure resources and workloads

Identity management features

Configure authentication for fleet workloads and users.

Google Cloud

And GKE expands beyond just Kubernetes! Yes, Kubernetes is at its core, but it incorporates other Google Cloud services and engineering practices to simplify multi-cluster and multi-cloud workloads.

Additional GKE services and tools that span beyond Kubernetes include:

Configuration and policy management tools to scale your workloads by automatically adding and updating identical configurations, features, and security policies across your fleet.

Team management tools provide team access and visibility to infrastructure resources and workloads.

Cloud Service Mesh provides powerful tools for application security, networking, and observability across your cluster infrastructure (often called a **mesh**) for microservice-based applications running in your fleet.

Identity management features to consistently configure authentication for fleet workloads and users.

And **observability features** to monitor and troubleshoot your clusters and applications, including their health, resource utilization, and security posture.

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Google Cloud

With fleets, you can view and control clusters from a single view in the Google Cloud console. But fleets are more than just groups of clusters. They rely on **sameness** and **trust** to provide a foundation for advanced multi-cluster management capabilities.

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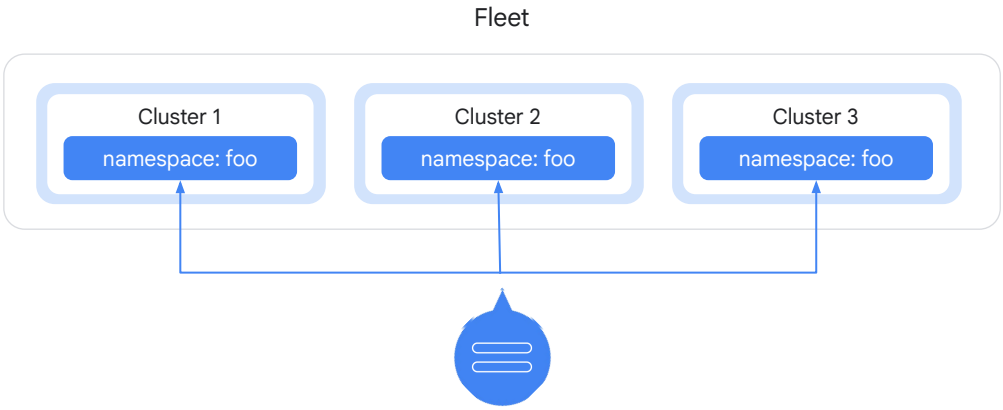
Sameness simplifies fleet management



Google Cloud

Within a fleet of clusters, some Kubernetes objects that have the same name, like namespaces, are treated equally, even if they are in different clusters. This is called **sameness**, and it simplifies fleet management.

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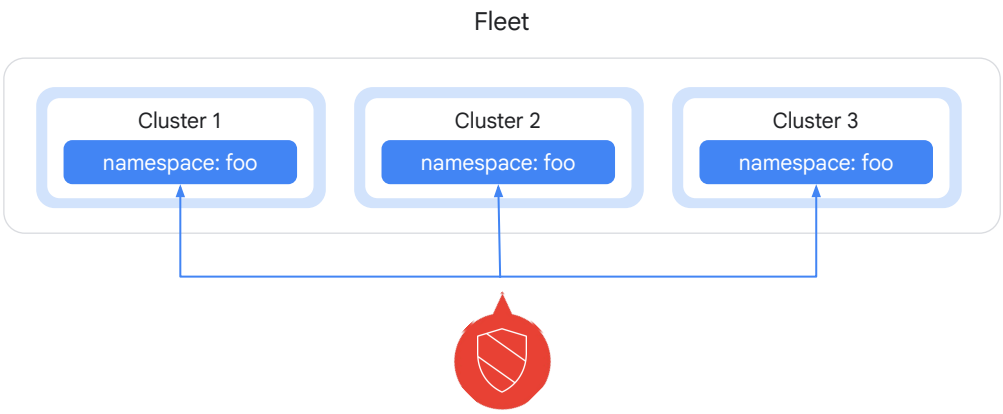
Use Policy Controller to apply a security policy



Google Cloud

For example, you can use Policy Controller to apply a security policy to all fleet services with the namespace ‘foo’, regardless of where the clusters are located.

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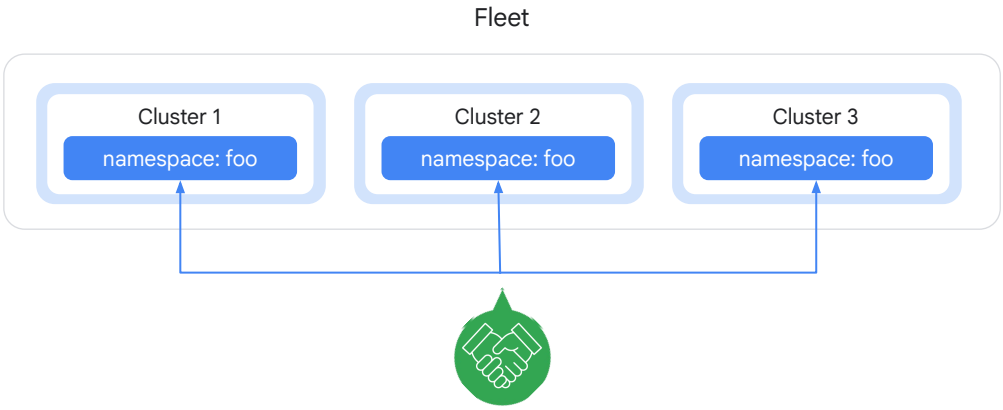
Use trust to manage resources at the fleet level



Google Cloud

Another key principle is **trust**. When clusters and teams trust each other, resources can be managed at the fleet level instead of individually configuring each cluster. This simplifies management and allows for consistent policies across all environments.

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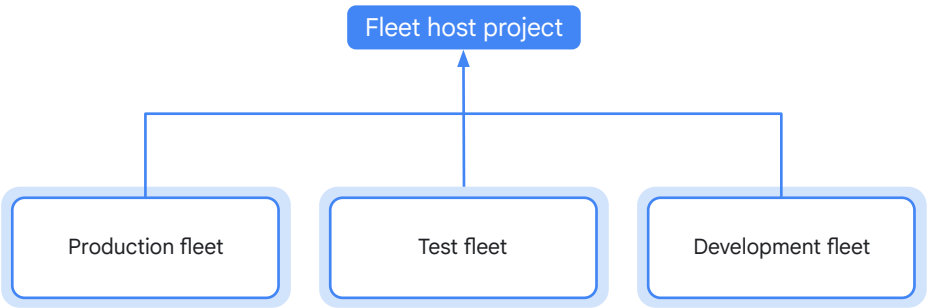
Organize fleets based on organizational requirements



Use **scopes** to Control access and manage teams.

Google Cloud

Organize your fleet based on your organization's unique technical and operational requirements.

Each fleet is associated with a Google Cloud project called the **fleet host project**. This project is primarily used to manage and view your fleet, but it can also include clusters from other projects.

For example, you can use fleets for your production, test, and development environments, or for different departments in your company. Within each fleet, you can use scopes to control access and manage different teams.

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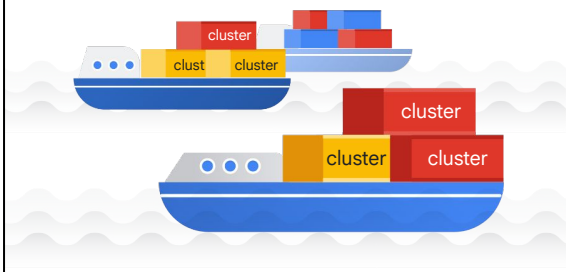
The advantages of managing clusters with fleets



- Manage clusters and services that frequently communicate.
- Improve resource management and security by aligning clusters with environments (for example, production, development, and other environments)
- Ensure scalability and simplified management by using the largest possible fleet size that preserves sameness and trust.

Google Cloud

Clusters with services that frequently communicate with each other have significant advantages when managed in a fleet.

To improve resource management and security, align your Kubernetes clusters with your environments. Create dedicated fleets for production, development, and other environments. This isolation enhances stability and allows for environment-specific optimizations.

Strive for the largest possible fleet size that preserves sameness and trust. This allows for scalability and easier management. When you need more granular access control, Cloud Service Mesh provides that flexibility.

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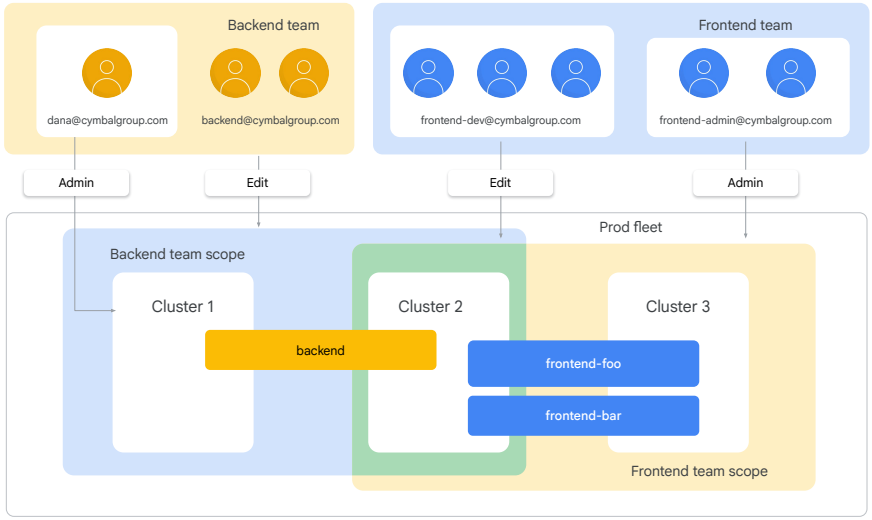
Teams in GKE



Google Cloud

We have mentioned Teams a couple of times in this module. A team is a way to organize and manage access to your fleet resources. On this slide we have 2 teams; Backend and Frontend. Each team has access to resources with the cluster, includes some resources that may be shared with other teams and some that are not.

Teams might go against the principles of **sameness** within GKE for several reasons, even though sameness is designed to simplify management and ensure consistency. Some reason to do this are:

1. Varying Application Requirements for teams
2. Development and Testing Differences
3. Organizational Silos and Autonomy
4. Compliance and Regulatory Requirements
5. Technical Debt and Legacy Systems
6. Lack of Understanding or Training
7. Resistance to Change

We will examine teams in more detail later in this course.

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Introduction to GKE at scale

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Google Cloud

GKE provides several features to help you effectively manage your fleets and applications.

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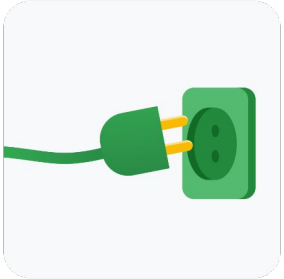
Connect Agent



Connect Agent

- A Kubernetes deployment used to manage fleet connections between Google Cloud and other cloud providers
- Automatically installed when you register your cluster, regardless of where the cluster is located
- Creates a secure connection between your cluster and your Google Cloud host project, regardless of where your cluster is located.

Google Cloud

Connect Agent is a Kubernetes deployment used to manage fleet connections between Google Cloud and other cloud providers.

When you register your cluster within a Google Cloud fleet, the connect agent is automatically installed. This agent creates a secure connection between your cluster and your Google Cloud host project, regardless of where the cluster is located. This lets you manage your clusters and workloads from Google Cloud and use Google Cloud services.

While beyond the scope of this course, GKE also provides support for connectivity to on-premises environments.

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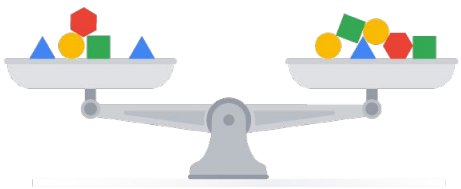
Load balancing solutions



Default load balancers:

Layer 4: External passthrough Network Load Balancers

Layer 7: External Application Load Balancers

- You can use Multi-Cluster Ingress to distribute traffic across multiple clusters with a single load balancer.
- There are modes available for load balancing on-premises clusters.
- Clusters hosted on public clouds use platform-native load balancers.

Google Cloud

You can manage network traffic across your fleet with built-in load balancing solutions in GKE, ensuring optimal performance and high availability.

The default GKE load balancers are external passthrough Network Load Balancers for Layer 4 and external Application Load Balancers for Layer 7. Both are managed services, so you don't need to worry about configuration or provisioning.

With Multi Cluster Ingress, you can easily distribute traffic across multiple clusters using a single load balancer.

GKE has modes for load balancing clusters hosted on premises. You can also manually configure load balancing to integrate with your existing solutions.

GKE clusters hosted on public clouds use platform-native load balancers.

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Authentication and authorization

- You can use your Google identity to authenticate to clusters across your fleet using Google credentials.
- Your teams can also use their existing login credentials from third-party providers to access any cluster in your fleet.
- Depending on the environment, you can also use SAML, LDAP.

Google Cloud

Authentication and authorization can be significant challenges when working with multiple clusters across multiple providers. GKE provides consistent, simple, and secure authentication solutions for interacting with your clusters.

Regardless of where your clusters are located, you can use your Google identity to authenticate to clusters across your fleet using Google credentials. Directly access your clusters or integrate your Google identity with automated tools like build pipelines and DevOps workflows.

With GKE, your teams can also use their existing login credentials from **third-party providers** like Microsoft Active Directory Federation Service, AzureAD, and Okta to access any cluster in your fleet.

Depending on the environment, you can also use SAML, LDAP.

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Policy Controller

- Consistent enforcement of security and regulatory compliance across a fleet can be challenging.
- Validates every API request to your Kubernetes cluster and ensures compliance.

Google Cloud

Another challenge when working with multiple clusters is consistent enforcement of security and regulatory compliance policies across a fleet. Many organizations have strict requirements. For example, financial service applications that need to protect consumer information. These organizations must meet these compliance standards at scale.

Policy Controller validates every API request to your Kubernetes cluster and ensures that they comply with your organization's specific security and governance rules.

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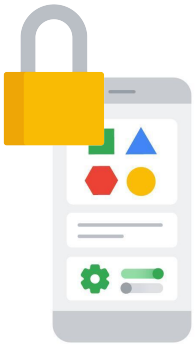
Application-level security



- Use binary authorization to ensure that only trusted images are deployed on your fleet's clusters.
- Network policy lets you specify which Pods are allowed to communicate with each other and other network endpoints.
- Cloud Service Mesh gives you fine-grained control over which services can communicate with each other.

Google Cloud

GKE also provides robust security, access control, and authentication features for applications.

Binary Authorization helps ensure that only trusted images are deployed on your fleet's clusters.

Kubernetes network policy lets you specify which Pods are allowed to communicate with each other and other network endpoints.
Cloud Service Mesh access control gives you fine-grained control over which services can communicate with each other. You can use service accounts and request attributes to define precise access rules for your applications.

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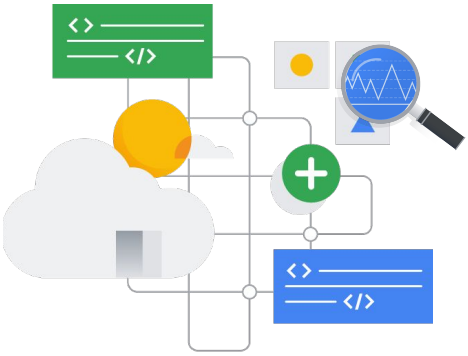
Observability



- GKE brings enterprise observability features and dashboards to the Google Cloud console.
- You can use Cloud Logging and Cloud Monitoring for more detailed information.
- GKE integrates with observability tools like Prometheus, Grafana, Elastic, and Splunk.
- Cloud Service mesh provides a unified control plane for microservices-based applications.

Google Cloud

The Google Cloud console is the command center for managing Google Cloud projects and resources. GKE brings enterprise observability features and dashboards to the Google Cloud console for full visibility into your fleet. View high-level details or drill into the data to identify issues and troubleshoot.

For more in-depth information about your clusters and their workloads, you can use Cloud Logging and Cloud Monitoring.

Depending on your organization and project needs, GKE also integrates with other observability tools, like open source Prometheus and Grafana, and third-party tools such as Elastic and Splunk.

For microservices-based applications, Cloud Service Mesh provides a unified control plane and integrates with existing tools for efficiency operation and troubleshooting at scale.